

CSE370 : Database Systems

Assignment 05/11/2025 | Fall 2025

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```
CREATE DATABASE Company_23101137;  
USE Company_23101137;
```

```
CREATE TABLE Employee
```

```
(  
    employee_id char(10),  
    first_name varchar(20),  
    last_name varchar(20),  
    email varchar(60),  
    phone_number char(14),  
    hire_date date,  
    job_id char(10),  
    salary int,  
    commission_pct decimal(5,3),  
    manager_id char(10),  
    department_id char(10)  
);
```

```
INSERT INTO Employee VALUES
```

```
('EMP001', 'Nafis', 'Rahman', 'nafis.rahman@company.com', '+8801712345678', '2024-01-15', 'JOB001', 25000, 0.050, 'MNG001', 'DPT001'),  
( 'EMP002', 'Tasneem', 'Hossain', 'tasneem.hossain@company.com', '+8801812345678', '2023-12-10', 'JOB002', 2000, 0.073, 'MNG002', 'DPT002'),  
( 'EMP003', 'Nahid', 'Hasan', 'nahid.hasan@company.com', '+8801912345678', '2024-02-20', 'JOB003', 27000, 0.064, 'MNG003', 'DPT003'),  
( 'EMP004', 'Arafat', 'Islam', 'arafat.islam@company.com', '+8801711111111', '2024-03-05', 'JOB004', 3200, 0.081, 'MNG004', 'DPT004'),  
( 'EMP005', 'Tasneem', 'Ahmed', 'tasneem.ahmed@company.com', '+8801811111111', '2024-01-30', 'JOB005', 2600, 0.054, 'MNG005', 'DPT005'),  
( 'EMP006', 'Muhtadi', 'Kamal', 'muhtadi.kamal@company.com', '+8801911111111', '2023-11-25', 'JOB006', 26000, 0.042, 'MNG006', 'DPT006'),
```

('EMP007', 'Farhana', 'Begum', 'farhana.begum@company.com', '+8801712222222', '2024-04-10', 'JOB007', 35000, 0.101, 'MNG007', 'DPT007'),
 ('EMP008', 'Naima', 'Chowdhury', 'naima.chowdhury@company.com', '+8801812222222', '2023-10-15', 'JOB008', 25000, 0.031, 'MNG001', 'DPT001'),
 ('EMP009', 'Zubair', 'Khan', 'zubair.khan@company.com', '+8801912222222', '2024-02-28', 'JOB009', 29000, 0.063, 'MNG002', 'DPT002'),
 ('EMP010', 'Ishrat', 'Jahan', 'ishrat.jahan@company.com', '+8801713333333', '2023-09-05', 'JOB003', 22000, 0.064, 'MNG003', 'DPT003'),
 ('EMP011', 'Mahmud', 'Hassan', 'mahmud.hassan@company.com', '+8801723456789', '2024-05-12', 'JOB011', 26700, 0.04, 'MNG004', 'DPT004'),
 ('EMP012', 'Sharmin', 'Akter', 'sharmin.akter@company.com', '+8801823456789', '2023-08-20', 'JOB011', 2800, 0.06, 'MNG005', 'DPT005'),
 ('EMP013', 'Fahim', 'Uddin', 'fahim.uddin@company.com', '+8801923456789', '2024-06-15', 'JOB012', 31000, 0.09, 'MNG006', 'DPT006'),
 ('EMP014', 'Sadiah', 'Rahman', 'sadia.rahman@company.com', '+8801724567890', '2023-07-25', 'JOB013', 23000, 0.03, 'MNG007', 'DPT007'),
 ('EMP015', 'Raihan', 'Kabir', 'raihan.kabir@company.com', '+8801824567890', '2024-05-05', 'JOB014', 2700, 0.07, 'MNG001', 'DPT001');

MariaDB [Company_23101137]> SELECT * FROM Employee;

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id
EMP001	Nafis	Rahman	nafis.rahman@company.com	+8801712345678	2024-01-15	JOB001	25000	0.050	MNG001	DPT001
EMP002	Tasneem	Hossain	tasneem.hossain@company.com	+8801812345678	2023-12-10	JOB002	2000	0.073	MNG002	DPT002
EMP003	Nahid	Hasan	nahid.hasan@company.com	+8801912345678	2024-02-20	JOB003	27000	0.064	MNG003	DPT003
EMP004	Arafat	Islam	arafat.islam@company.com	+8801711111111	2024-03-05	JOB004	3200	0.081	MNG004	DPT004
EMP005	Tasneem	Ahmed	tasneem.ahmed@company.com	+8801811111111	2024-01-30	JOB005	2600	0.054	MNG005	DPT005
EMP006	Muhtadi	Kamal	muhtadi.kamal@company.com	+8801911111111	2023-11-25	JOB006	26000	0.042	MNG006	DPT006
EMP007	Farhana	Begum	farhana.begum@company.com	+8801712222222	2024-04-10	JOB007	35000	0.101	MNG007	DPT007
EMP008	Naima	Chowdhury	naima.chowdhury@company.com	+8801812222222	2023-10-15	JOB008	25000	0.031	MNG001	DPT001
EMP009	Zubair	Khan	zubair.khan@company.com	+8801912222222	2024-02-28	JOB009	29000	0.063	MNG002	DPT002
EMP010	Ishrat	Jahan	ishrat.jahan@company.com	+8801713333333	2023-09-05	JOB003	22000	0.064	MNG003	DPT003
EMP011	Mahmud	Hassan	mahmud.hassan@company.com	+8801723456789	2024-05-12	JOB011	26700	0.040	MNG004	DPT004
EMP012	Sharmin	Akter	sharmin.akter@company.com	+8801823456789	2023-08-20	JOB011	2800	0.060	MNG005	DPT005
EMP013	Fahim	Uddin	fahim.uddin@company.com	+8801923456789	2024-06-15	JOB012	31000	0.090	MNG006	DPT006
EMP014	Sadia	Rahman	sadia.rahman@company.com	+8801724567890	2023-07-25	JOB013	23000	0.030	MNG007	DPT007
EMP015	Raihan	Kabir	raihan.kabir@company.com	+8801824567890	2024-05-05	JOB014	2700	0.070	MNG001	DPT001

15 rows in set (0.000 sec)

No 1 Query (as Plain Text)	SELECT first_name, last_name, email, phone_number, hire_date, department_id FROM Employee WHERE hire_date = (SELECT MAX(hire_date) FROM Employee);
No 1 SS (of Query & Output in Shell)	<pre> MariaDB [Company_23101137]> SELECT first_name, last_name, email, phone_number, hire_date, department_id -> FROM Employee -> WHERE hire_date = (SELECT MAX(hire_date) FROM Employee); +-----+-----+-----+-----+-----+-----+ first_name last_name email phone_number hire_date department_id +-----+-----+-----+-----+-----+-----+ Fahim Uddin fahim.uddin@company.com +8801923456789 2024-06-15 DPT006 +-----+-----+-----+-----+-----+-----+ 1 row in set (0.000 sec) </pre>
No 2 Query (as Plain Text)	SELECT e.first_name, e.last_name, e.employee_id, e.phone_number, e.salary, e.department_id FROM Employee e WHERE e.salary = (SELECT MIN(salary) FROM Employee e2 WHERE e2.department_id = e.department_id)

No 2 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23101137]> SELECT e.first_name, e.last_name, e.employee_id, e.phone_number, e.salary, e.department_id
-> FROM Employee e
-> WHERE e.salary = (
->   SELECT MIN(salary) FROM Employee e2
->   WHERE e2.department_id = e.department_id
-> );
```

first_name	last_name	employee_id	phone_number	salary	department_id
Tasneem	Hossain	EMP002	+8801812345678	2000	DPT002
Arafat	Islam	EMP004	+8801711111111	3200	DPT004
Tasneem	Ahmed	EMP005	+8801811111111	2600	DPT005
Muhtadi	Kamal	EMP006	+8801911111111	26000	DPT006
Ishrat	Jahan	EMP010	+8801713333333	22000	DPT003
Sadia	Rahman	EMP014	+8801724567890	23000	DPT007
Raihan	Kabir	EMP015	+8801824567890	2700	DPT001

```
7 rows in set (0.000 sec)
```

No 3 Query
(as Plain Text)

```
SELECT first_name, last_name, employee_id, commission_pct, department_id FROM Employee WHERE department_id =
"DPT007" AND commission_pct < ALL(SELECT commission_pct FROM Employee WHERE department_id =
"DPT005");
```

No 3 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23101137]> SELECT first_name, last_name, employee_id, commission_pct, department_id FROM Employee WHERE department_id = "DPT007" AND
commission_pct < ALL(SELECT commission_pct FROM Employee WHERE department_id = "DPT005");
```

first_name	last_name	employee_id	commission_pct	department_id
Sadia	Rahman	EMP014	0.030	DPT007

```
1 row in set (0.000 sec)
```

No 4 Query
(as Plain Text)

```
SELECT department_id, COUNT(employee_id) AS total_employees
FROM Employee
GROUP BY department_id
HAVING MAX(salary) <= 30000;
```

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [company_23101137]> SELECT department_id, COUNT(employee_id) AS total_employees
-> FROM Employee
-> GROUP BY department_id
-> HAVING MAX(salary) <= 30000;
+-----+-----+
| department_id | total_employees |
+-----+-----+
| DPT001        | 3               |
| DPT002        | 2               |
| DPT003        | 2               |
| DPT004        | 2               |
| DPT005        | 2               |
+-----+-----+
5 rows in set (0.000 sec)
```

No 5 Query
(as Plain Text)

```
SELECT DISTINCT e.department_id, e.job_id, e.commission_pct
FROM Employee e
WHERE EXISTS (
  SELECT * FROM Employee e2
  WHERE e2.department_id = e.department_id
  AND e2.job_id != e.job_id
  AND e2.commission_pct > e.commission_pct
);
```

No 5 SS
(of Query & Output
in Shell)

```
MariaDB [company_23101137]> SELECT DISTINCT e.department_id, e.job_id, e.commission_pct  
-> FROM Employee e  
-> WHERE EXISTS (  
->   SELECT * FROM Employee e2  
->   WHERE e2.department_id = e.department_id  
->     AND e2.job_id != e.job_id  
->     AND e2.commission_pct > e.commission_pct  
-> );
```

department_id	job_id	commission_pct
DPT001	JOB008	0.031
DPT002	JOB009	0.063
DPT004	JOB011	0.040
DPT007	JOB013	0.030
DPT005	JOB005	0.054
DPT006	JOB006	0.042
DPT001	JOB001	0.050

7 rows in set (0.000 sec)

No 6 Query
(as Plain Text)

```
SELECT DISTINCT m.manager_id  
FROM Employee m  
WHERE m.manager_id IS NOT NULL  
AND NOT EXISTS (  
  SELECT * FROM Employee e  
  WHERE e.manager_id = m.manager_id  
  AND e.salary < 3500  
);
```

No 6 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23101137]> SELECT DISTINCT m.manager_id
-> FROM Employee m
-> WHERE m.manager_id IS NOT NULL
-> AND NOT EXISTS (
-> SELECT * FROM Employee e
-> WHERE e.manager_id = m.manager_id
-> AND e.salary < 3500
-> );
```

```
+-----+
| manager_id |
+-----+
| MNG003     |
| MNG006     |
| MNG007     |
+-----+
```

```
3 rows in set (0.000 sec)
```

No 7 Query
(as Plain Text)

```
SELECT first_name, last_name, employee_id, email, salary, department_id, commission_pct
FROM Employee e
WHERE e.manager_id IS NOT NULL
AND e.commission_pct = (
SELECT MIN(e2.commission_pct)
FROM Employee e2
WHERE e2.manager_id = e.manager_id
);
```

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [Company_23101137]> SELECT first_name, last_name, employee_id, email, salary, department_id, commission_pct  
-> FROM Employee e  
-> WHERE e.manager_id IS NOT NULL  
->    AND e.commission_pct = (  
->    SELECT MIN(e2.commission_pct)  
->    FROM Employee e2  
->    WHERE e2.manager_id = e.manager_id  
-> );
```

first_name	last_name	employee_id	email	salary	department_id	commission_pct
Nahid	Hasan	EMP003	nahid.hasan@company.com	27000	DPT003	0.064
Tasneem	Ahmed	EMP005	tasneem.ahmed@company.com	2600	DPT005	0.054
Muhtadi	Kamal	EMP006	muhtadi.kamal@company.com	26000	DPT006	0.042
Naima	Chowdhury	EMP008	naima.chowdhury@company.com	25000	DPT001	0.031
Zubair	Khan	EMP009	zubair.khan@company.com	29000	DPT002	0.063
Ishrat	Jahan	EMP010	ishrat.jahan@company.com	22000	DPT003	0.064
Mahmud	Hassan	EMP011	mahmud.hassan@company.com	26700	DPT004	0.040
Sadia	Rahman	EMP014	sadia.rahman@company.com	23000	DPT007	0.030

8 rows in set (0.000 sec)